

SYSTEMS PROJECT MANAGEMENT

Systems Analysis & Design

Author: Eng. Carlos Andrés Sierra, M.Sc.
cavirguezs@udistrital.edu.co

Full-time Adjunct Professor
Computer Engineering Program
School of Engineering
Universidad Distrital Francisco José de Caldas

2026-I



UNIVERSIDAD DISTRITAL
FRANCISCO JOSÉ DE CALDAS

Outline

1 Information Systems

2 Enterprises as Systems

3 Software Methodologies



Outline

- 1 Information Systems
- 2 Enterprises as Systems
- 3 Software Methodologies



Information Systems

- An **Information System** is a system that *collects, processes, stores,* and *disseminates* information.
 - Information systems are used to support and manage business operations.
 - Information systems are used to automate and optimize business processes.
 - Examples of information systems include transaction processing systems, management information systems, decision support systems, executive information systems, expert systems, and data systems.
- Handwritten notes:*
- management
 - decisions
 - predictions (ML)
 - business intelligence (dashboards + descriptive)



Information Systems

- An **Information System** is a **system** that *collects, processes, stores, and disseminates* information.
 - **Information systems** are used to **support** and **manage** business operations. $\Rightarrow$ **business value**
 - **Information systems** are used to **automate** and **optimize** business processes.
 - Examples of information systems include transaction processing systems, management information systems, decision support systems, executive information systems, expert systems, and data systems.
- Handwritten notes:* Red arrows point from "support" and "manage" to "business value". Red arrows point from "automate" and "optimize" to "Engineering".



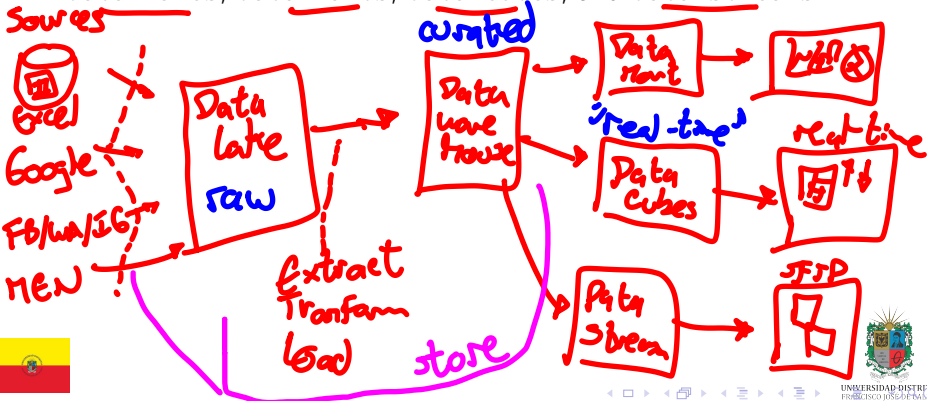
Information Systems

- An **Information System** is a **system** that *collects, processes, stores, and disseminates* information.
- **Information systems** are used to **support** and **manage** business operations.
- **Information systems** are used to **automate** and **optimize** business processes.
- Examples of **information systems** include **transaction processing systems**, **management information systems**, **decision support systems**, **executive information systems**, **expert systems**, and **data systems**.



Data Systems

- A Data System is a system that collects, processes, stores, and retrieves data.
- Data systems are used to store and analyze data.
- Examples of data systems include databases, data warehouses, data lakes, data marts, data cubes, and data streams.



Expert Systems

- An **Expert System** is a system that uses knowledge and reasoning to solve problems. **90's**
- **Expert systems** are used to automate and optimize decision-making processes.
- Examples of expert systems include diagnostic systems, predictive systems, prescriptive systems, decision support systems, and automated reasoning systems. **Rules**



Expert Systems

- An **Expert System** is a system that uses **knowledge** and **reasoning** to solve problems.
- **Expert systems** are used to **automate** and **optimize** decision-making processes.
- Examples of **expert systems** include diagnostic systems, predictive systems, prescriptive systems, decision support systems, and automated reasoning systems.

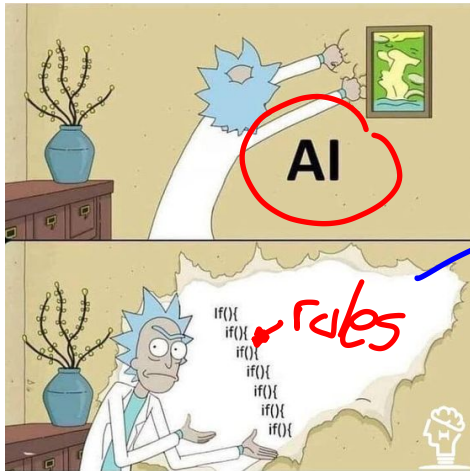
→ regression

↓
chatbot 270



Expert Systems as Classical Artificial Intelligence

Here is a great example of a diagnostic system.



decision
tree



Risks and Failures in Information

- **Information systems** are subject to **risks** and **failures** that can **impact** business operations.
- **Risks and failures** can be mitigated through **security measures**, **backup systems**, **disaster recovery plans**, and **monitoring tools**.
- Examples of risks and failures include security breaches, data loss, system downtime, performance issues, and compliance violations.

protocols



Risks and Failures in Information

- **Information systems** are subject to **risks** and **failures** that can impact business operations.
- **Risks and failures** can be mitigated through **security measures**, **backup systems**, **disaster recovery plans**, and **monitoring tools**.
- Examples of **risks and failures** include security breaches, data loss, system downtime, performance issues, and compliance violations.

rules



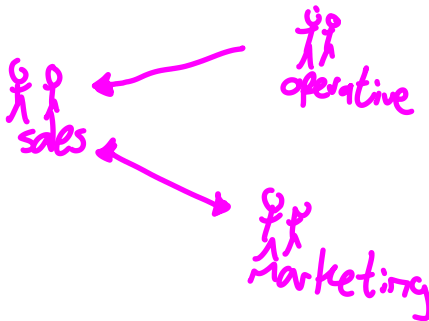
Outline

- 1 Information Systems
- 2 Enterprises as Systems
- 3 Software Methodologies



Enterprises: Bottom-Up and Top-Down Approaches

- **Bottom-Up Approach:** Analyzes an enterprise by examining its individual units or components, then aggregating them to understand the entire organization.
- **Top-Down Approach:** Starts with an overall vision or strategy and decomposes it into subsystems, departments, and processes.

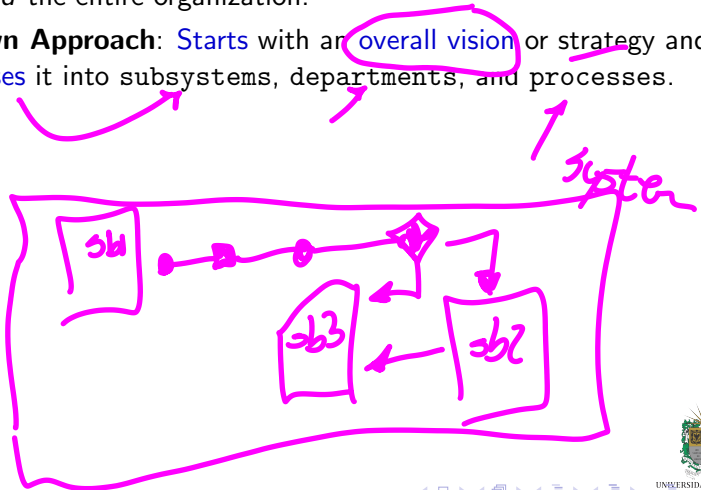


emerge
behavior



Enterprises: Bottom-Up and Top-Down Approaches

- **Bottom-Up Approach:** Analyzes an enterprise by examining its individual units or components, then aggregating them to understand the entire organization.
- **Top-Down Approach:** Starts with an overall vision or strategy and decomposes it into subsystems, departments, and processes.



PIECE Framework for Enterprises

- **P**articipation: *Engaging stakeholders* at every level. *thinking*
- **I**ndependence of Thought: Encouraging *diverse* and *innovative ideas*.
- **E**laboration: Developing and *refining ideas* and *processes*.
- **C**ommunication: Ensuring clear and *effective exchange* of information. *improve*
- **E**xploration: Embracing *continuous innovation* and *improvement*.

Helps → growth / steady state



Enterprise System Typologies

- **Rational Systems:** Organizations driven by logical and structured processes with clear hierarchies.
- **Natural Systems:** Organizations viewed as organic entities with informal structures and emergent behavior.
- **Open Systems:** Enterprises that continuously interact with their external environment for information, resources, and feedback.



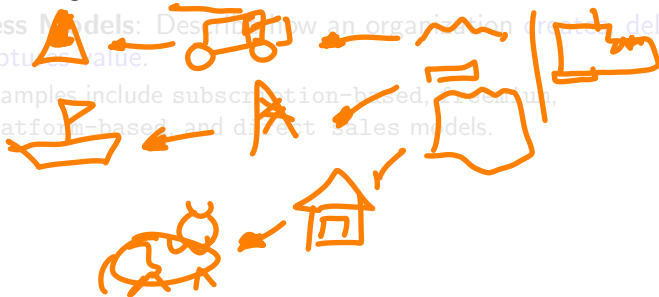
information
systems

Human



Business Systems and Models

- **Business Systems:** Integrated frameworks that encompass an enterprise's processes, operations, and strategies.
- **Examples:** Enterprise Resource Planning (ERP) systems, Customer Relationship Management (CRM) systems, Supply Chain Management (SCM) systems.
- **Business Models:** Describe how an organisation creates, delivers, and captures value.
 - Examples include subscription-based, freemium, platform-based and direct sales models.



Business Systems and Models

- **Business Systems:** Integrated frameworks that encompass an enterprise's processes, operations, and strategies.
- **Examples:** Enterprise Resource Planning (ERP) systems, Customer Relationship Management (CRM) systems, Supply Chain Management (SCM) systems. ERP
- **Business Models:** Describe how an organization creates, delivers, and captures value. Netflix youtube spotify
 - Examples include subscription-based, freemium, platform-based, and direct sales models.

Cloud
Computing

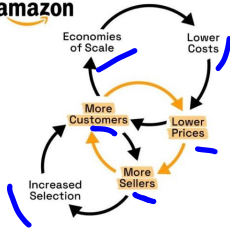
products



Business Models Examples

Understanding Business Models Through Flywheels

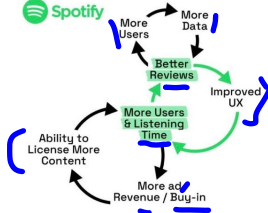
amazon



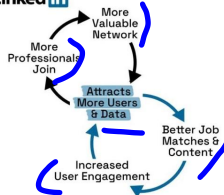
NETFLIX



Spotify



LinkedIn



Outline

- 1 Information Systems
- 2 Enterprises as Systems
- 3 Software Methodologies

gradient

$$\begin{bmatrix} \frac{\partial z}{\partial x} & \frac{\partial z}{\partial y} \end{bmatrix}$$

z

x

y



Basic Concepts in Software Methodologies

- **Software methodologies** provide frameworks for planning, designing, developing, testing, and maintaining software projects.
- They help teams manage project complexity and ensure quality deliverables.

6 months → min
3-6 people
8 h/d

system



Traditional Methodologies

- **Waterfall** A linear sequential approach where each phase must be completed before moving to the next.

- Suitable for projects with well-defined requirements and *minimal uncertainty*.

- Emphasizes thorough *documentation* and up-front planning.

Projects → small specific money

Requirements

↳ Analysis (system)

↳ Design

↳ Implementation (development coding)

↳ Test

↳ Deployment



Traditional Methodologies

- **Waterfall:** A linear sequential approach where each phase must be completed before moving to the next.
- Suitable for projects with well-defined requirements and minimal uncertainty.
- Emphasizes thorough documentation and up-front planning.

80's
evolution

• risks
• iteration

all
no internet
no standards
no frameworks

no changes



Agile Methodologies

\$ money ↓ | "Friends" 3-4

- Emphasizes iterative development, customer collaboration, and adaptability to change. MVP
- Based on the Agile Manifesto (2001), which values individuals and interactions over processes and tools. natural system

1991 → Internet ☺

80s → PCs — nachiskash
— Alca
— Ion

↓
less
cost ↑

new
business

↓
new applications

- uncertainty model
- no final business



Agile Methodologies

- Emphasizes iterative development, customer collaboration, and adaptability to change.
- Based on the **Agile Manifesto** (2001), which values individuals and interactions over processes and tools.
- Examples include Scrum, Kanban, Extreme Programming (XP), and Lean Software Development.
- Agile methodologies are suitable for projects with evolving requirements and high uncertainty.
- Promotes adaptive planning, evolutionary development, and early delivery of working software.

→ rugby

1995 → tech ~ 2008



Agile Methodologies

- Emphasizes **iterative development**, **customer collaboration**, and **adaptability to change**.
- Based on the **Agile Manifesto** (2001), which values **individuals and interactions** over processes and tools.
- *Examples* include Scrum, Kanban, Extreme Programming (XP), and Lean Software Development.
- **Agile methodologies** are suitable for projects with **evolving requirements** and **high uncertainty**.
- Promote **adaptive planning**, **evolutionary development**, and **early delivery** of working software.

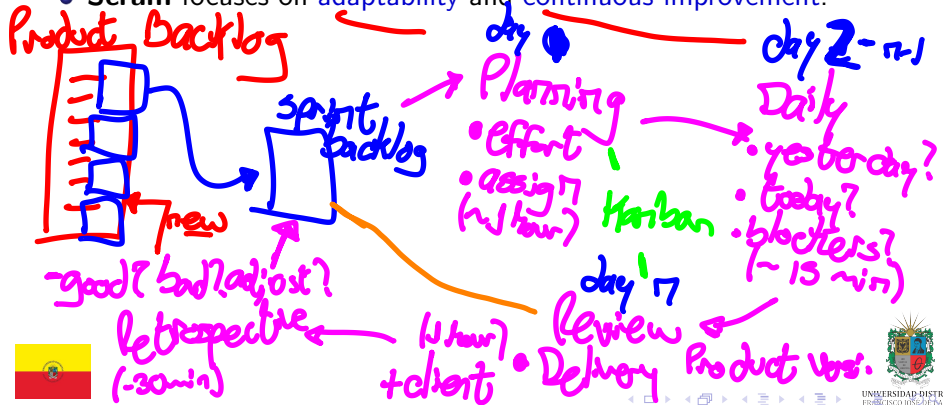
MVP (1.0) (Beta)



Case Study: Scrum

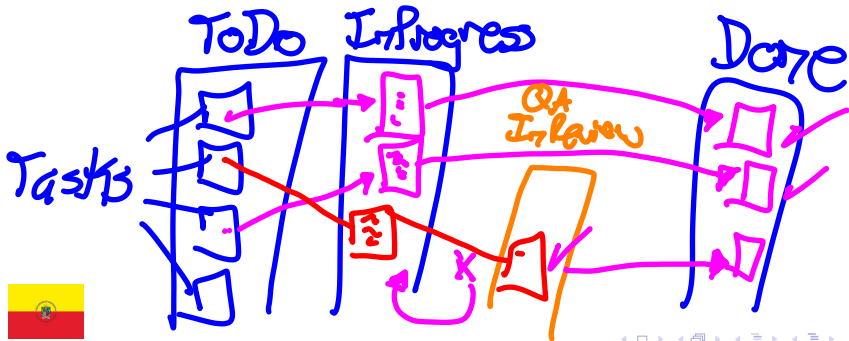
~2 weeks → 1 week

- **Scrum** employs short, time-boxed iterations called sprints (typically 1-4 weeks). *→ core no viej*
- Key events include daily stand-ups, sprint planning, sprint reviews, and sprint retrospectives.
- **Scrum** focuses on adaptability and continuous improvement.



Case Study: Kanban

- **Kanban** visualizes work flow on boards and limits Work In Progress (WIP) to optimize flow.
- Emphasizes continuous improvement, flow optimization, and just-in-time delivery.
- Ideal for support teams and maintenance projects requiring *continuous flow* with minimal iteration planning.



Outline

- 1 Information Systems
- 2 Enterprises as Systems
- 3 Software Methodologies



Thanks!

Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/systems-analysis>

